

Homework 9

Instructions:

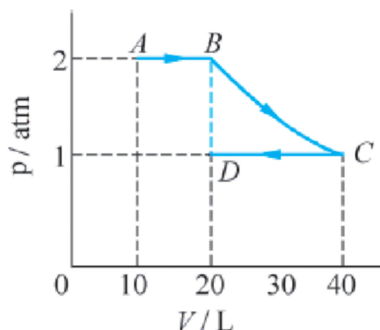
- Problems marked with an asterisk (*) are optional and not graded. They are for students who would like extra practice

This assignment is not required for submission and is meant for your own review and practice. If you choose to submit it, please upload your work to Web Learning by June 10.

1. Thermodynamic Process on a $p - V$ Diagram

A fixed mass of an ideal gas undergoes a state change in the direction indicated by the arrows on the curve shown in the figure. Section BC of the curve is a hyperbola with the p -axis and V -axis as its asymptotes.

- Given that the temperature of the gas at state A is $T_A = 300K$, find the temperatures of the gas at states B , C , and D .
- What is the total work done by the gas on its surroundings from state A to state D ?
- Sketch this entire process on a $V - T$ diagram, clearly indicating the direction of the process.



2. Adiabatic and Isothermal Processes

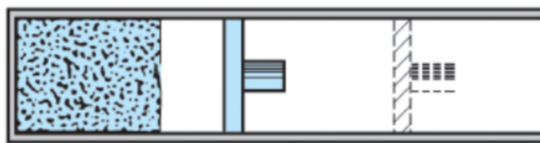
A sample of n moles of an ideal gas (with adiabatic index γ) initially has a pressure p_1 and a volume V_1 . It is first adiabatically compressed to a volume V_2 (where $V_2 < V_1$), and then allowed to expand isothermally back to its original volume V_1 .

- Find the maximum pressure and the maximum temperature reached during this entire process.
- Find the heat absorbed by the gas, the work done by the gas on its surroundings, and the change in its internal energy during this entire process.
- Sketch the curves for the entire process on a $p - V$ diagram.

3. Free Expansion and Controlled Expansion

As shown in the figure, a cylinder is constructed with insulated walls and an insulated piston. Initially, the total volume of the cylinder is V_0 . A partition divides it into two parts: one part with a volume V_1 contains an ideal nitrogen gas of mass m (molar mass M) at a pressure p_1 , while the other part is a vacuum. A small hole in the partition is now opened, allowing the gas to fill the entire cylinder. Then, the piston is slowly moved to let the gas expand until its final volume reaches V_2 .

- Find the final pressure and temperature of the gas.
- Find the total work done by the gas on its surroundings and the change in its internal energy during the entire process as its volume changes from V_1 to V_2 .



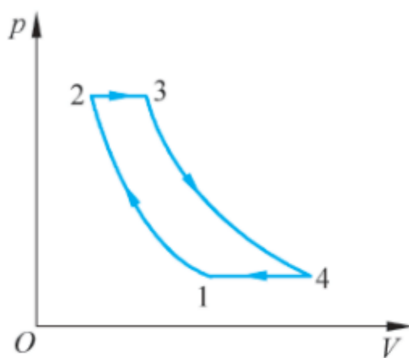
4. Efficiency of a Brayton Cycle

The cyclic process carried out by gas turbines used in modern jet aircraft and thermal power plants can be simplified as the Brayton cycle shown in the figure. Process $1 \rightarrow 2$ represents a fixed mass of air being adiabatically compressed into the combustion chamber; $2 \rightarrow 3$ represents fuel being injected and burned in the combustion chamber, resulting in an isobaric expansion of the gas; $3 \rightarrow 4$ represents the high-temperature and high-pressure gas being directed into a turbine, expanding adiabatically to drive the turbine blades and do work; $4 \rightarrow 1$ represents the waste gas entering a heat exchanger where it is compressed isobarically, releasing heat to a coolant (air or water).

- Prove that the efficiency of this cycle expressed in terms of the temperatures at points 1, 2, 3, and 4 is:

$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

- Let $r_p = p_{\max}/p_{\min}$ represent the pressure ratio of this cycle, which allows the efficiency to be expressed as $\eta = 1 - \frac{1}{r_p^{(\gamma-1)/\gamma}}$. If the $\gamma = 1.40$ and the pressure ratio is $r_p = 10$, what is the efficiency of the engine?



5. Ocean Thermal Energy Conversion

It is possible to construct a heat engine that utilizes the temperature difference between surface seawater and deep seawater. Suppose the surface water temperature in a tropical ocean region is T_H and the deep water temperature is T_C (both expressed in Kelvin).

- What is the efficiency of a Carnot engine operating between these two temperatures?
- If a power plant operates at this maximum theoretical efficiency and obtains a net mechanical power output of P , at what rate (power) will it discard waste heat to the surroundings?
- Both the mechanical power obtained and the discarded waste heat come from the heat released by cooling the surface water down to the deep water temperature. At what rate (mass flow rate $\frac{dm}{dt}$) must the power plant draw surface seawater? Assume the specific heat capacity of water is a constant c .

6. Entropy Changes via Multi-stage Thermal Reservoirs

- A mass m of water at an initial temperature T_1 is placed in contact with a constant-temperature thermal reservoir at T_2 ($T_1 < T_2$) and eventually reaches thermal equilibrium. Find the total change in entropy of the system composed of the water and the thermal reservoir. Does the total entropy increase or decrease? Assume the specific heat capacity of water is a constant c .
- If the same water of mass m and initial temperature T_1 is first placed in contact with an intermediate constant-temperature reservoir at T_m ($T_1 < T_m < T_2$) until equilibrium is reached, and then moved to the thermal reservoir at T_2 until it reaches equilibrium again, find the total entropy change of the entire system (the water and both thermal reservoirs) and compare your result with that obtained in part (a).

7. Entropy of Mixing for Liquids

Inside an insulated container, two portions of the same liquid are mixed under constant pressure. The masses of the two portions are equal, both being m , but their initial temperatures are different, being T_1 and T_2 respectively, with $T_2 > T_1$. The two portions reach a new equilibrium state after mixing. Find the total change in entropy of the system caused by this mixing, and prove that the total entropy increases. The specific heat capacity at constant pressure c_p is a constant.

8*. Characterization of Polytropic Processes

A thermodynamic process of an ideal gas that is neither isothermal nor adiabatic, but can be expressed by the process equation $pV^n = \text{constant}$, is called a polytropic process, where n is the polytropic index.

- Explain what types of thermodynamic processes correspond to $n = 0, 1, \gamma$, and ∞ .
- Prove that the work done by the surroundings on the ideal gas during a polytropic process is given by:

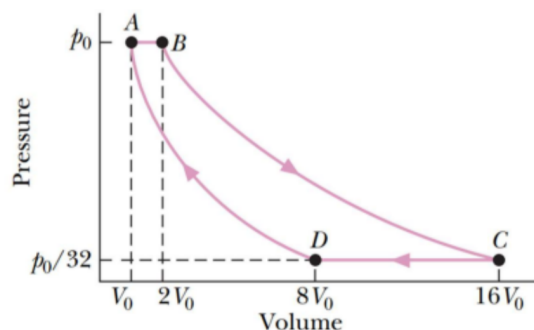
$$W = \frac{p_2 V_2 - p_1 V_1}{n - 1}$$

- Prove that the molar heat capacity of an ideal gas in a polytropic process is given by $C_m = C_{V,m} \left(\frac{\gamma - n}{1 - n} \right)$, and use this relation to state the value of C_m for each process identified in part (a).

9*. Engine Cycle Analysis

An ideal gas (1.00 mol) is the working substance in an engine that operates on the cycle shown in the figure. Processes BC and DA are reversible and adiabatic. The volume parameters are given as $V_A = V_0$, $V_B = 2V_0$, $V_D = 8V_0$, and $V_C = 16V_0$, and the pressure parameters are $p_A = p_B = p_0$ and $p_C = p_D = p_0/32$.

- Is the gas monatomic, diatomic, or polyatomic?
- What is the thermal efficiency of this engine?



10*. **Entropy of Mixing for Gases at Different Pressures**

Two insulated containers each contain ν moles of the same ideal gas. Initially, the two containers are isolated from each other, but they are at the same temperature while their initial pressures are p_1 and p_2 , respectively. The two containers are then connected so that the gas eventually reaches a new equilibrium state. Prove that the total entropy change of the system caused by this process is:

$$\Delta S = \nu R \ln \frac{(p_1 + p_2)^2}{4p_1 p_2}$$

Furthermore, prove that $\Delta S > 0$.